

Figure 12.8 Finding a quote using a positional query

This example shows why `span` queries come in handy. They are low-level queries that help us find documents with tokens specified by positions. When working with legal documents, research articles, or technical books, where sentences with the words' exact positions are required, we can use `span` queries. There are several kinds of `span` queries—`span_first`, `span_within`, `span_near`, and so on—and we take a peek at a few of them here; as always, refer to the documentation to learn more.

12.8.1 Sample data

Before we work with `span` queries, let's prime Elasticsearch with a `quotes` index and a couple of documents.

Listing 12.14 Priming Elasticsearch for a `span` query

```
PUT quotes                          <- Creates a quotes index with
{                                      a couple of properties
  "mappings": {
    "properties": {
      "author":{
        "type": "text"
      },
      "quote":{
        "type": "text"
      }
    }
  }
}

                                    <- Indexes
PUT quotes/_doc/1                      Newton's quote
{
  "author":"Isaac Newton",
  "quote":"Plato is my friend. Aristotle is my friend.
  ➥ But my greatest friend is the truth."
}
```

We create the `quotes` index with a couple of properties: `author` and `quote`, which are both text fields (`"type": "text"`). We also index the quote by Isaac Newton as defined in the listing. Now that we have primed the `quotes` index, let's see some `span` queries in action, beginning with the `span_first` query.